Assignment 4 - Refined methods and preliminary research results

Submission Date: Sunday, October 25th, 2026, 11:59pm

Class presentations: Friday, October 30th, 2026, 11am, with David Attaway of AWS

In this assignment you should build on the research proposal developed in Assignment 3. Start by refining your selected research question and method. Then obtain some associated data and run a small but meaningful pilot analysis to generate some prototype results. The aim is to demonstrate that the proposed work is plausible and that you can make progress toward the final research paper.

Create a 4-minute progress presentation for the 11am class on Friday, October 30th, with David Attaway of AWS. The presentation should include a problem, a suitable research question, a method and some preliminary results, along with expected the results or the coursework project by the end of the semester.

Ideally you can show 1-2 substantive results as polished graphics (synthetic data is fine as long as you identify it). Also include other things to bring your presentation to life, such as a study area map or descriptive data plots, or some visualization of pilot model performance. A small, well-checked analysis is sufficient. And it helps to be transparent about limitations and the work needed before drawing stronger conclusions.

Table 1 Research progress aspects to address.

Component	Question to address
Research question and progress	What question are you pursuing and what have you refined since Assignment 3? What could be the overall contribution of the work?
Refined method	Specify the study area, unit of analysis, data preparation, analytical steps, agent roles, human decisions, etc. Include a workflow diagram, a baseline method, verification/validation checks and relevant spatial assumptions.
Data and pilot implementation	Show that you have accessed and inspected usable data and executed a meaningful part of the workflow. Report your sources, coverage, sample size, key quality issues, and the code or tools used.
Preliminary results and graphics	Show evidence from your pilot analysis, ideally as 1-2 clear graphics. State what was measured, how it compares with your baseline. Also, comment on your overall project feasibility. Label preliminary findings.
Limitations and next steps	Explain what remains untested, any failed checks or changes needed, and a realistic plan to complete the paper. Identify one focused question to gain feedback from the class.

Please submit your presentation online as a PDF by 11:59pm on Sunday, October 25th. You will present it at the 11am class on Friday, October 30th. The submitted slides must already include your refined method and preliminary evidence. You may improve the graphics and extend the analysis before presenting, clearly identify any results added or changed since submission.

Prepare professional, legible slides with consistent formatting, slide numbers, source acknowledgments and an AI disclaimer identifying the tools used, their contributions and your own checks. Use supporting slides for full references, methodological details and evidence files. Proofread and rehearse a clear four-minute explanation.

Treat this as a professional research briefing. Give each graphic a clear takeaway, readable labels, units, legends and data sources as appropriate. Explain what the result shows and why it matters to your question. Be prepared to discuss the method, the evidence and your next steps with our guest.

Table 2 Assessment criteria.

Criterion	Evidence	Weight
Refined research method	Clear question and contribution. Justified improvements since Assignment 3. Explicit data, workflow, baseline and verification/validation, etc.	30%
Preliminary evidence and feasibility	Usable data and an executed pilot, with informative results and graphics. You have checks that support a credible assessment of feasibility and limitations.	30%
Critical human review	Documented decisions about AI outputs, data and code checks, honest interpretation of preliminary findings, and a realistic plan for what to do next.	20%
Presentation quality and delivery	Professional graphics and slides. Proper citations with a suitable AI usage statement. Clear and well presented oral explanation, which keeps to time.	20%

Students remain responsible for the accuracy and defensibility of all submitted material. Verify citations and any AI-generated code. Distinguish observed pilot results from expected outcomes and unresolved claims. Remember, agreement between agents is not independent scientific validation.